How-To Library

Informatica

Performance Tuning Guidelines for Lookup Transformations in Streaming Mappings

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Abstract

When you use Lookup transformations in Streaming mappings, multiple factors impact mapping performance. You can optimize performance when you configure Lookup transformation properties and execution parameters in the Streaming mapping. This article provides recommendations that you can use to optimize mapping performance.

Supported Versions

- Informatica Intelligent Streaming 10.1.1

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Overview

Use Informatica Intelligent Streaming mappings to collect streaming data, build the business logic for the data, and push the logic to a Spark engine for processing. The Spark engine uses Spark Streaming to process data. The Spark engine reads the data, divides the data into micro batches, and publishes it.

You can use Lookup transformations in Streaming mappings to look up data in a flat file, HDFS, Hive, or Sqoop.

You can improve the performance of a Lookup transformation by configuring the following properties:

- Lookup transformation advanced properties
- Execution parameters in streaming mapping properties

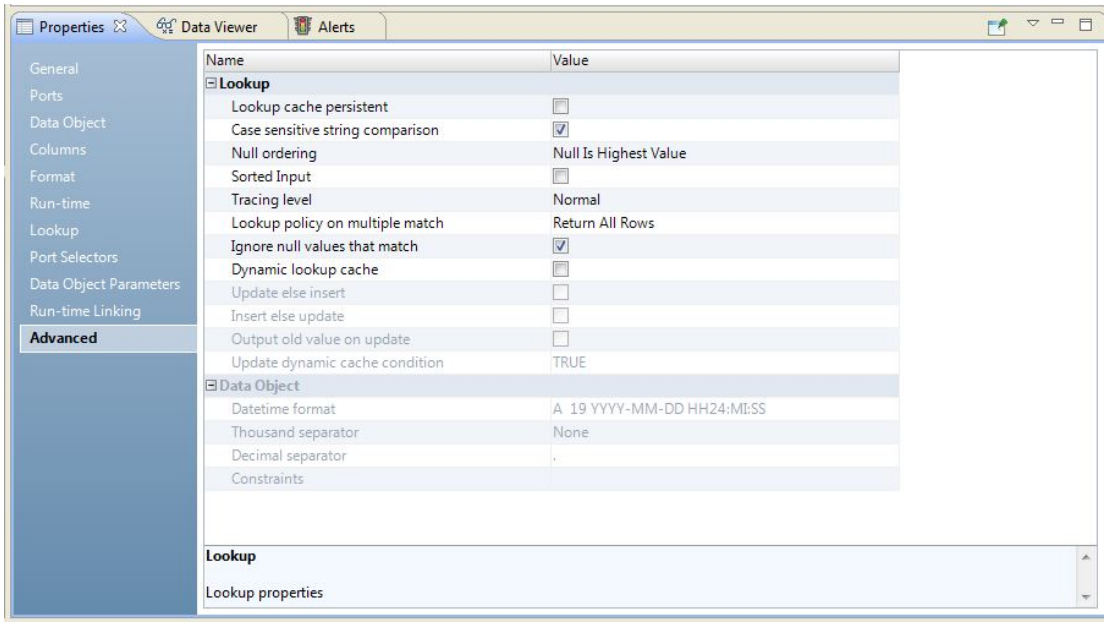
For more information about streaming mappings and transformations used in the streaming mappings, see the Informatica Intelligent Streaming User Guide.

Lookup Transformation Advanced Properties

To optimize the performance of the Lookup transformation, enable the **Ignore null values that match** option on the **Advanced Properties** tab of the Lookup transformation.

When you enable this option, Intelligent Streaming ignores the null value comparison in the lookup condition and Spark does not do a cross join (Cartesian product) of DataFrames. If you enable this option, Spark does a sort-merge-join for equality joins.

The following image shows the **Ignore null values that match** option:

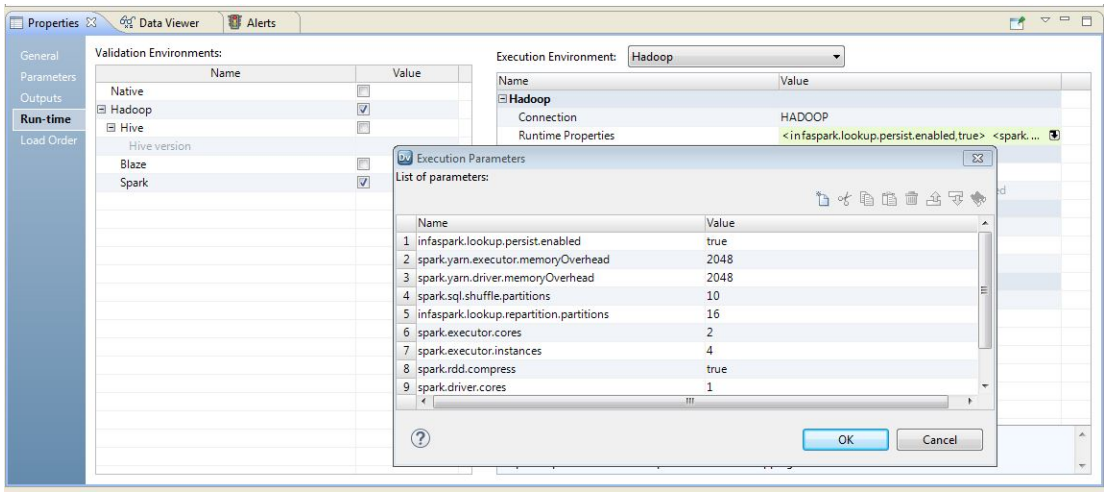


Execution Parameters

When you use a Lookup transformation, you can configure optional Streaming mapping properties to improve the performance of the transformation. Specify these properties in the execution parameters of the Streaming mapping Run-time properties and in the custom properties for the Data Integration Service.

When you want Spark to cache the lookup data, set the `infaspark.lookup.persist.enabled` parameter to true. The data is then read from the cache in Spark instead of the source, for instance a Hive source. This is an optional execution parameter that you can configure if the data that the source data object is reading is large and the lookup stage is taking a long time.

The following image shows the execution parameters that you can configure as part of the Streaming mapping Run-time properties:



If you set the `infaspark.lookup.persist.enabled` parameter to true, configure the following parameters:

spark.streaming.unpersist

To prevent Spark Streaming from unpersisting the persisted RDDs, you must set the `spark.streaming.unpersist` execution parameter to false

Infaspark.lookup.repartition.partitions

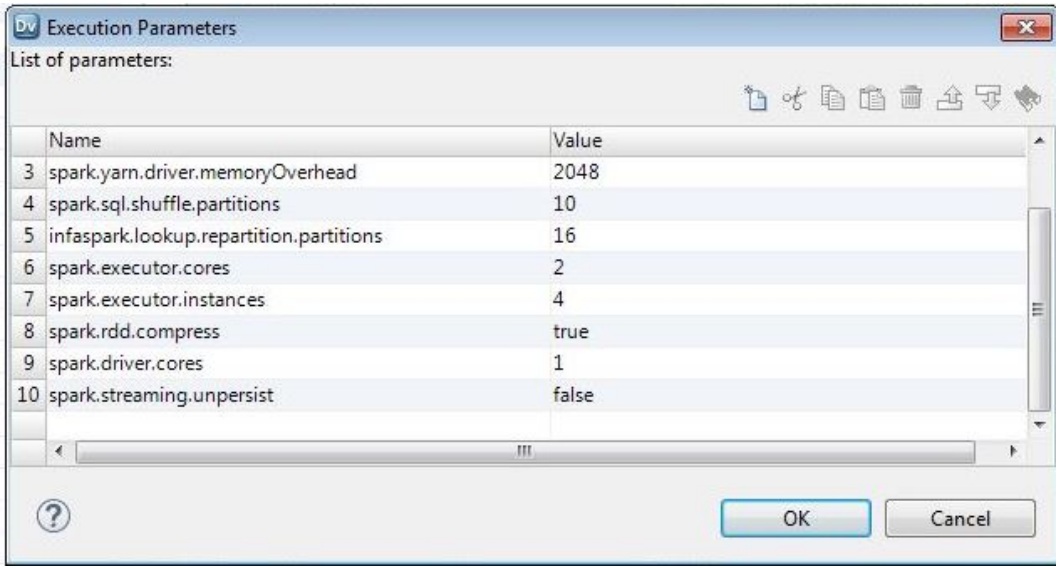
Specifies the number of partitions that Intelligent Streaming repartitions the Lookup RDDs into. The number of partitions impacts the total amount of available parallelism.

Optionally, to increase or decrease the number partitions in the RDD, specify a value for the `Infaspark.lookup.repartition.partitions` execution parameter.

spark.rdd.compress

Specifies whether to compress the serialized partitions in the RDD. This can save substantial space, but at the cost of CPU time. Optionally, to compress the serialized partitions, set the `spark.rdd.compress` execution parameter to true.

The following image shows the `spark.streaming.unpersist`, `Infaspark.lookup.repartition.partitions`, and the `spark.rdd.compress` execution parameters:



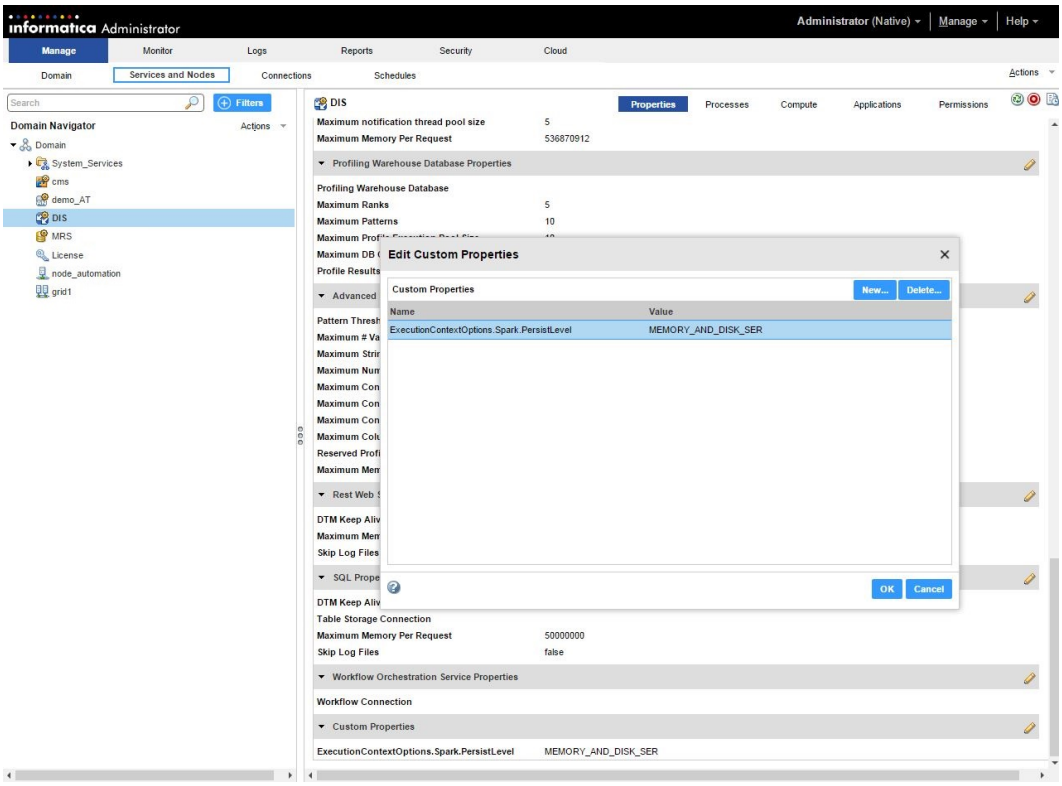
ExecutionContextOptions.Spark.PersistLevel

Specifies the storage level at which the RDD is persisted. The default storage level is `MEMORY_AND_DISK`.

To modify the storage level, configure this property in the custom properties of the Data Integration Service. You configure the Data Integration Service properties in the Administrator tool. You can specify one of the following values for the property:

- DISK_ONLY
- MEMORY_ONLY
- MEMORY_ONLY_SER
- MEMORY_AND_DISK
- MEMORY_AND_DISK_SER

The following image shows the Data Integration Service custom properties:



Authors

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